Lecture 5: Vector spaces & subspaces.

Main examples: $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \dots, \mathbb{R}^n, \dots$

"Definition": A vector space is a set of vectors equipped with rules of
- addition
- multiplication by real numbers.

In a vector space,
- addition satisfies
$$x+y = y+x$$
- there is a vector 0 such $x+0=x$ for all x
- there is a vector "$-x$" such that $x+(-x)=0$.
- ...

Examples:
1) $\mathbb{R}^\infty \ni (1,2,-1,3,\dots)$
2) Space of 2×3 matrices.
   $\to$ Looks like $\mathbb{R}^6$
3) Space of functions $f(x)$ (defined for $0 \leq x \leq 1$, say).
   $f(x) = \sin(x) \quad g(x) = 3x^2$
   $\Rightarrow (f+g)(x) = \sin(x) + 3x^2$

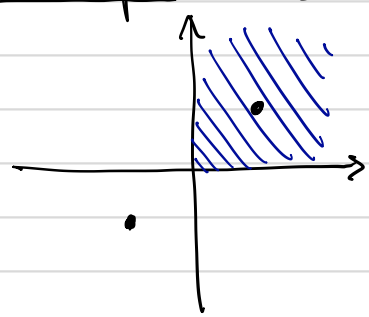
(+) 4) Any plane thru $\begin{bmatrix}0\\0\\0\end{bmatrix} \in \mathbb{R}^3$.

Definition: V vector space.
A subspace of V is a (nonempty) subset $S \subset V$ satisfying:
1) If $x, y \in S$, then $x+y \in S$
2) If $x \in S$, $c \in \mathbb{R}$, then $cx \in S$.

Note: The zero vector $0 \in V$ belongs to S. (Rule 2: take $c=0$.)

Example: Subspaces of $\mathbb{R}^3$:
- $\{0\} \in \mathbb{R}^3$
- lines thru $0 \in \mathbb{R}^3$
- planes thru $0 \in \mathbb{R}^3$
- $\mathbb{R}^3$

Not example: First quadrant $\subset \mathbb{R}^2$
$S = \{x \geq 0, y \geq 0\}$



$\begin{bmatrix}1\\1\end{bmatrix} \in S$, but $-\begin{bmatrix}1\\1\end{bmatrix} \notin S$

S not subspace: disrespects rule 2!